

Introduction To Object-Oriented Programming

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What We'll Learn

By the end of this lesson, students can:

- Explain the basic idea of Object-Oriented Programming.
- Identify objects, classes, attributes, and methods.
- Describe the relationship between a class and an object.
- Identify real-world examples that can be represented as Java objects.

Before We Start

Look at a real-world object, such as a student, cellphone, car, or computer.

Think about the following questions:

- What information describes the object?
- What actions can the object perform?
- How can we represent the object in a Java program?

Let's Understand

Object-Oriented Programming, or OOP, is a way of organizing a program using objects.

An object represents something that contains information and can perform actions. It may represent a real-world item, person, place, or idea.

Examples of objects include:

- A student
- A teacher
- A cellphone
- A bank account
- A school subject

Classes

A class is a blueprint or template used to create objects.

For example, a Student class describes the information and actions that every student object may have.

The class itself is not a specific student. It only defines what student objects should contain and what they should be able to do.

Objects

An object is a specific instance created from a class.

For example, if Student is a class, the following can be objects created from it:

- Angela
- Joshua
- Maria

Each student object can store different information even though they were created from the same class.

Attributes

Attributes are variables that store information about an object. They describe the object's characteristics or current state.

A student object may have the following attributes:

- Name
- Age
- Grade level
- Section

A cellphone object may have the following attributes:

- Brand
- Model
- Color
- Battery level

Methods

Methods represent the actions or behaviors that an object can perform.

A student object may have the following methods:

- Study
- Submit an activity
- Take an examination
- Display student information

A cellphone object may have the following methods:

- Make a call
- Send a message
- Take a picture
- Charge the battery

Class And Object Relationship

A class defines the structure and behavior of objects.

An object is created based on that class.

For example:

- Class: Student
- Object: student1
- Attributes: name, age, and section
- Methods: study() and displayInformation()

One class can be used to create multiple objects. Each object can have its own attribute values.